



SPE 94583

First Applications of a Multiple Fracturing Method in Noncemented Horizontal Offshore Wells

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This paper was prepared for presentation at the SPE Latin American and Caribbean Petroleum Engineering Conference held in Rio de Janeiro, Brazil, 20 – 23 June 2005.

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Abstract

This paper presents the first offshore applications, both in platform and subsea wells, of a multiple fracture creation process in openhole horizontal wells, previously applied in onshore scenarios. The applications had to overcome many challenges typical of workovers on mature field offshore wells. The paper covers detailed workover planning, multiple fracturing designs, operations data, and the stimulation results. The campaign comprised three horizontal slotted liner wells offshore Brazil as part of a research project to evaluate selective stimulation methods for horizontal wells. It was proved that the method can be successfully applied in platform and subsea wells (a first). The economic results were very attractive, making the method an alternative for hundreds of noncemented offshore horizontal wells.

Introduction

The great success of horizontal wells offshore Brazil made them a basic case for every offshore project. The usual completion for horizontal wells in Brazil has been openhole gravel pack in friable sandstone, with some stand-alone screens cases, and preperforated liners in stable sandstone and carbonates. Cased cemented completions have not been used in horizontal wells offshore Brazil. The assumption that long horizontal sections eliminate the need for stimulation treatments, usually required with vertical well completion, has justified the noncemented horizontal completions¹. In the last few years, the operator has faced the challenges of underachieving horizontal wells, particularly rapid declining production. This has been a worldwide problem attributed to permeability anisotropy (particularly vertical permeability restriction), skin damage, near wellbore plugging of natural fracture network or ineffective stimulation¹⁻⁶. The last is particularly true in openhole horizontal wells and even more

so in slotted or preperforated completions⁷. Matrix treatments have not presented sufficient results, which posed a challenge for the operator to find technologies that could bypass damage and stimulate horizontal wells. Ideally, the creation of a limited number of discrete fractures, with proper length and conductivity, widely separated and well distributed, can achieve this goal⁸. There are many techniques to create these fractures in cemented horizontal wells. Whenever return on investment justifies the business case for a huge investment in casing and cementing of the horizontal well, then perforating and fracturing the well selectively, it should be applied. In regard to noncemented wells the placement of multiple fractures only where they are needed is a big issue⁸. To select the most reliable and economical techniques to stimulate horizontal wells in all these scenarios the operator launched a research project included as part of an Advanced Oil Recovery Program. In its search for alternative methods for new horizontal wells, the project started with a pilot well onshore and currently is fracturing selectively the first cemented horizontal well offshore Brazil. Concerning mature wells completed with screens or preperforated liners, this paper presents the planning, execution, and results of this research project line.

Project Management Organization and Approaches

The operator's exploration and production (E&P) organization comprises business units and respective assets, a service unit, a research center and some support services. The research center and well engineering group sponsored and coordinated this project. During the planning stage, teamwork was assigned with a local or application coordinator in the business units involved. Local coordination is essential, especially in offshore mature wells, which present numerous operational problems. Service companies and other partners were involved from the beginning. A *drilling the well on paper* approach was adopted to anticipate problems and generate contingency plans. The project adopted the usual approach of initiating the implementation of new technologies onshore or on fixed platforms to finally, apply them in deepwater subsea wells. Therefore, in the first phase two platform wells were hydraulically fractured with the selected technology and in the second and last phase a subsea well was acid fractured with the same technology.

The Differentials of Mature Offshore Wells

There are many differences between offshore wells and land wells regarding costs, logistics, safety, and environmental concerns, and application techniques. The offshore logistics are more complex, more weather-dependent and much more expensive. Safety and environmental protection concerns are also more stringent. In deep-water subsea wells the application techniques become much more complex; in fact, those that are extremely time-consuming become cost-prohibitive. The introduction of a new technology applied onshore in a subsea well often demands modification to help ensure its feasibility. When submarine blowout preventers (BOPs) are connected to the rig floor by drilling riser, floating rigs movement and increased operational complexity require special solutions. Compared to new wells, mature wells may present additional difficulties due to wear and tear, function of age, materials used, and fluids produced, and sometimes old-fashioned technologies applied, etc. These differentials may be observed in low quality preperforated liners of platforms wells (stimulated in the first phase) or in a tubing head recall that had to be attended in the subsea well (in the second project phase). It is also common in mature well interventions to take on associated or extra operations, which extend the intervention and increase its complexity. This occurred in the subsea well in which an exploratory evaluation in a shallower zone increased the complexity of the stimulation.

Selecting Technologies for Multiple Fracturing of Noncemented Horizontal Wells

Successful placement of fractures in a horizontal section depends on the wellbore design, ability to isolate desired fractures zones, wellbore orientation to the fracture plane, and achievable pump rates⁸. A literature review carried out by the project team^{1,8-11}, revealed that conventional techniques do not allow the creation of distinct separated fractures at the desired depths in noncemented horizontal wells. Techniques, such as high-rate water fracturing, have produced poor results as observed in well production profiling and fracture-mapping techniques^{8,12,13}. Perforating at selected intervals to enhance the distribution of fractures along the horizontal section has produced limited success. The creation of slots using hydrajetting rather than perforating to create specific zones of weakness presented a little more success. The use of sealing devices, for example inflatable packers, tends to be useless in noncemented horizontal wells because fractures often jump over them, generating a passageway. An emergent technology, HydraJet Fracturing, was selected as a promising solution, successfully applied onshore, to create multiple fractures in noncemented horizontal wells offshore Brazil. The technique is a combination of three distinct processes: i) hydrajetting, ii) fracturing (down the tubing), and iii) co-injection down the annulus⁷. Rather than using mechanical sealing or chemical blocks, the HydraJet Fracturing process uses the fluids own dynamic to divert the majority of the fluid flow into a specific point in the horizontal section. This hydraulic sealing effect is based on the Bernoulli equation and is illustrated in **Fig 1**. As pressurized fluid exits the jetting tool, the pressure energy is transformed into kinetic energy, or velocity. When the fluid velocity around the jet stream is the highest, pressure in this area is the lowest. Hence, the fluids pumped down the annular

and even from fractures previously created are suctioned to the jetting point.

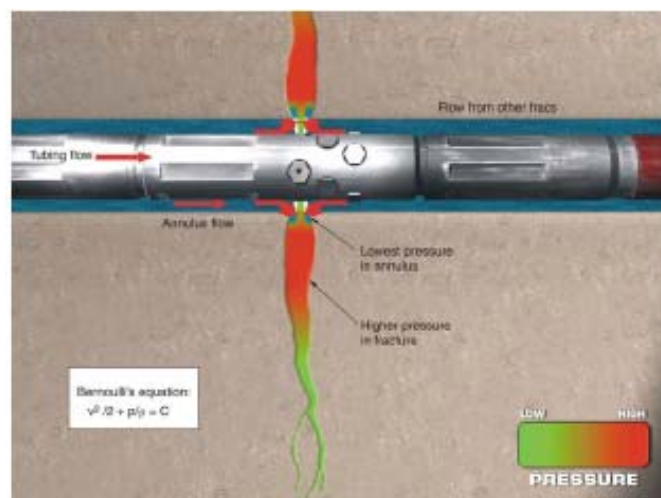


Fig 1. Dynamic sealing by hydrajet fracturing

The HydraJet Fracturing process allows the placement of many fractures of different sizes, sequentially, from the toe to the heel of the horizontal section. Fracture positioning can be very precise depending on the positioning tools that are used. The process allows the application of hydraulic fracturing, acidized fracturing or mixed systems at different locations along the horizontal section^{8,14}. Hydraulic fracturing was applied in the platform wells and acidized fracturing in the subsea well. As many texts^{3,14-17} thoroughly described the HydraJet Fracturing process, this paper will not discuss it in detail. Anyway, the item HydraJet Fracturing Execution presents some operation details.

Project Phase 1: Two Platform Wells

We present here a brief summary of this phase, as other paper⁷ described it in detail. The team recognized two underperforming noncemented horizontal wells drilled in the late 1990 in the Camarin field offshore northeast Brazil as good candidates for multiple fracturing. These wells penetrate horizontally a thin (less than 15 m) sandstone layer with low horizontal permeability (11 mD) and very low vertical permeability (0.000445 mD). Large fracturing is the only choice for stimulating this formation. Three fractures were created in the first well. In the first fracture at 3050 m MD a total of 51,400 lb of 18/30 bauxite and 7,500 lb of 16/30 RCP were pumped into the fracture. In the second fracture at 2900 m MD a total of 54,000 lb of 18/30 bauxite and 3,750 lb of 16/30 RCP were placed into the fracture. Finally, in the third fracture at 2660 m MD similar volumes of proppant were pumped into the fracture. Some operational abnormalities were observed, including a stuck pipe immediately after the first fracturing. In the second well two fractures, instead of the four planned, were created. Two fracturing treatments were aborted due to the risk of stuck pipe and shared resources compromise. The following key aspects differentiated this application:

- i) These were a first application of HydraJet Fracturing offshore, so the first step in a learning curve;

- ii) The stimulation vessel used was a improvised one and not a dedicated top of line stimulation vessel;
- iii) The slotted liner completion was a low quality one regarding the slots control; there were evidence of formation sand wash-in during the operations;
- iv) The formation was severely depleted.

Despite these difficulties, this first stage, completed on May 2004, achieved its main goals: acquire knowledge to apply this emergent technology in deepwater subsea wells and increase production of the Camorim field. Particular adverse aspects of the 5in slotted liner completion configuration mostly caused the stuck pipe abnormalities in these operations.

Project Phase 2: Hydrajet Fracturing in a Deepwater Subsea Well in Campos Basin

The assignment of a multidisciplinary project team started this application on June 2003.

The first step was the selection of the best candidate for the first subsea application of the hydraJet fracturing process. A very low recovery factor carbonate, Macaé/Quissamã, in the Congo field was selected, since conventional stimulations had not produced the desired results. The Macaé/Quissamã in the Congo field produces from zone QM-20 since 11/21/1998 via a unique well, 1-RJS-512HA, completed in its horizontal section (602 m) with 7in slotted liner (Fig. 2).

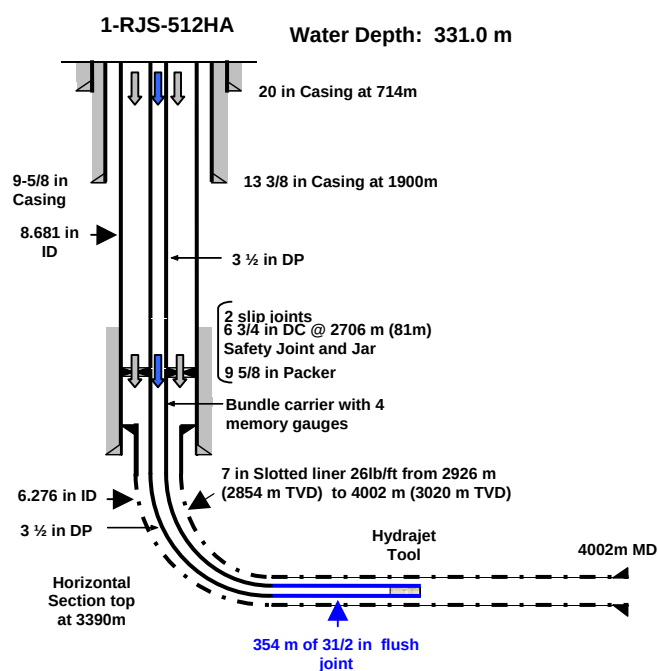


Fig 2. Wellbore schematics with hydrajet fracturing string

The second step was to determine the fracture direction around the well. From this point on a synergistic work between operator and service company reservoir and stimulation, people grew up. A geologic analysis of the faults in the area (Fig. 3) suggested that the fracture direction was most likely northwest-southeast. However, a recent study in Enchova field and a previous study in Bonito field, both carbonates near the Congo field, indicated almost east-west as

the fracture plane orientation. Anyway, there was doubt about the probable fracture plane orientation.

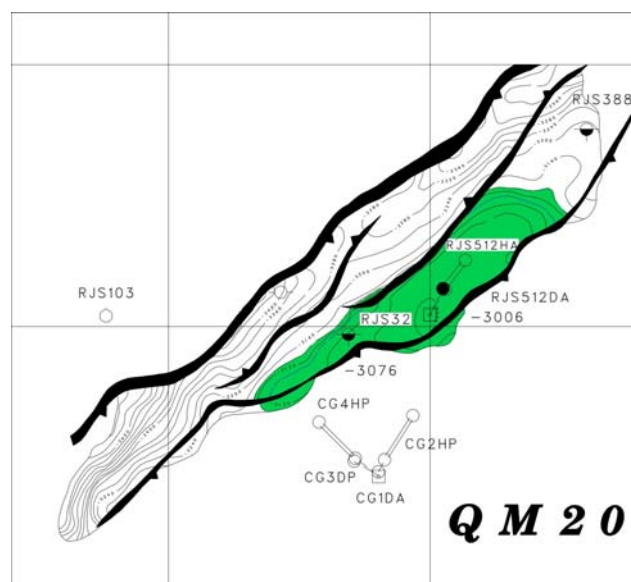


Fig 3. Topographical map – Congo QM-20 field

The third step was to define the type of stimulation to be applied. Table 1 shows the essential data to the well. Note that there were uncertainties about the reservoir properties and no data about rock mechanics and in-situ stresses.

Table 1 – Well and reservoir data – 1-RJS-512HA

Well name	1-RJS-512HA (subsea))
Water depth	331 m
Offshore Rig	SS-45 (moored floating rig)
Location	Congo, Campos Basin
Formation	Macaé/Quissamã 20
Lithology	Carbonate
Production slotted liner	7" 26# P-110 @ 2926 m
Production casing	9 5/8 in 47# P-110 @ 2972 m
Pore pressure	3569 psi @ 3050 m TVD
Bottom hole temperature	210 °F
Fluid type / API	Oil / 28 API
Fluid viscosity	1.007 @ 3000 psi (PVT)
Formation thickness	99 m gross / 28 m net
Avg Permeability	1 mD radial – 0,5 mD vertical
Avg Porosity	19 – 29%
Avg Water saturation	0.1%
Hydrajet tool	4.5 in OD w/8 jets 1/4 in

Examining this list and the previous information, one can infer the following:

- Low vertical permeability means that fracture stimulation would be ideal for this well.
- Slotted liner makes hydrajet fracturing the only reliable technique for the well
- A probably competent carbonate rock makes acidized fracturing a good option.
- Hydraulic fracturing was considered risky for this well based on the stuck pipe problems observed in the Camorim field.

The fourth step was to define the number and positions of the multiple fractures. A broad reservoir analysis (well test, logs, nodal, production history), a quick fracture design and the use of a numerical simulator to predict production after multiple fracture creation indicated that five fractures would be more adequate considering technological, logistical and economic aspects (**Fig. 4**). The simulation assumed $kh=1$, $kv=0.5$ and skin factor $S=25$ because these data provided a good production history match. It also assumed a 50 ft fracture length and 2,500 md-ft conductivity. The first four fractures were acid fractured and the fifth (near the heel) was supposed to be hydraulic (propped) fractured. The team decided to create at least one propped fracture as a feasibility test. However, as discussed later the fifth fracture was also acid fractured. Considering the horizontal section homogeneity, the fractures were almost evenly spaced at 3915, 3790, 3690, 3580 and 3435 m (MD).

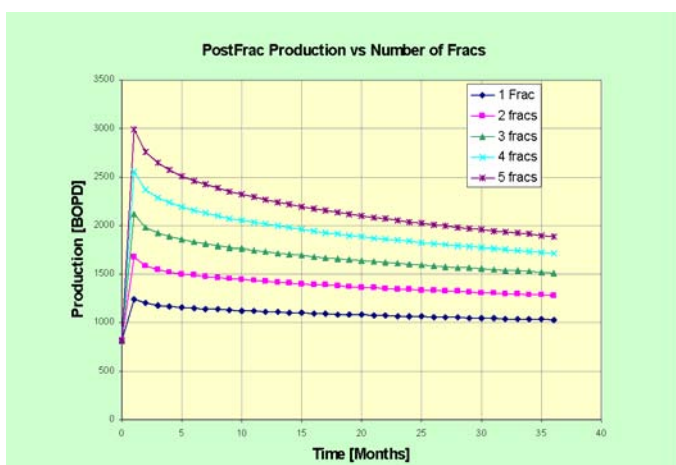


Fig 4. Production versus number of fractures

The fifth step was to write an intervention program detailing all the operations, in particular the hydrjet fracturing. From this point on a series of meeting, involving representatives of all the interfaces, including rig personnel was locally coordinated. The overall intervention was very complex as the stimulation was associated with a tubing head recall and an exploratory evaluation in a shallower zone. The tubing head substitution itself was a very challenging after it was determined that the regular approach was cost-prohibitive. The evaluation program in the shallower zone was also a big issue because it could damage the formation after its stimulation.

The Hydrjet Fracturing Design. Focusing only on Hydrjet Fracturing this application presented the following difficulties:

- The presence of two laterals, 9-RJS-512DA (a pilot well) and 1-RJS-512HP (abandoned due to a long fish in the horizontal section), near the main well, 1-RJS-512HA (**Fig. 5**). All the planned fractures could intersect the 1-RJS-512HP experiencing high additional leak off that could defer fracture growth. Note that the long fish (horizontal BHA) did not allow placing a cement plug along the open hole section.

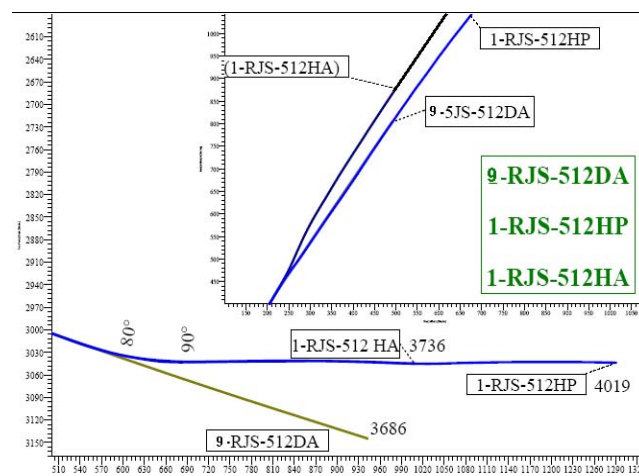


Fig 5. 9-RJS-512DA, 1-RJS-512HP and 1-RJS-512HA trajectories

- Keeping the hydrjet tool in position during each fracturing stage was initially a big issue in wells connected to floating rigs.
- Fluid loss control during fracturing while pulling the fracturing string out of the hole and when tripping the production tail string in hole was a significant concern.
- Pumping stimulation fluids under high pressure simultaneously down the tubing and the annulus from the stimulation vessel and the floating rig required modification of the onshore procedures.

The first two items were addressed in the hydrjet fracturing string specification (see below).

Fluid loss-control during the intervention was simplified, adopting the technique of feeding the well with completion fluid at only 7 bbl/hr rate. Fig 6 illustrates how the submarine BOP pumped simultaneously down both the operation string (3 ½ in DP) and the annulus different fluids.

Hydrjet Fracturing String Specification. The following needs guided the string specification: (1) proper jetting tool for four acid fracturings and one hydraulic fracturing; (2) sufficient flow rate capacity to pump fluids down the tubing and down the annulus considering extra leakoff due to multilateral wells; (3) means to maintain the hydrjetting tool in position during each fracturing; and (4) preventive measures concerning fishing job.

A 4 ½ in OD jetting tool with 8 x ¼ in jets and ball seat above and below it was selected. The eight jets would allow pumping 400,000 lb of proppant considering jets erosion limits.

The estimated treatment rates were 17-22 bbl/min down the tubing with 10,000-psi maximum pressure at surface. Assuming a 0.6 psi/ft fracture gradient, the estimated annular pressure at surface to extend the fracture was 1700 psi.

Initial computations showed that the annulus rate to extend the fracture was 4.5 bbl/min. Earlier experiences showed an increase of 1.5 bbl/min for each subsequent fracture placement. This means that during the fifth fracturing an annular rate of 10.5 bbl/min could be expected. Adding a possible double leak off (9.0 instead of 4.5 bbl/min) during

each fracturing, due to the laterals, the annulus rate could reach 15.0 bbl/min ($9.0 + 4 \times 1.5$ bbl/min). To provide room for a possible high annular rate the string OD should be a minimum. These considerations led the choice of a 3.5 in DP string.

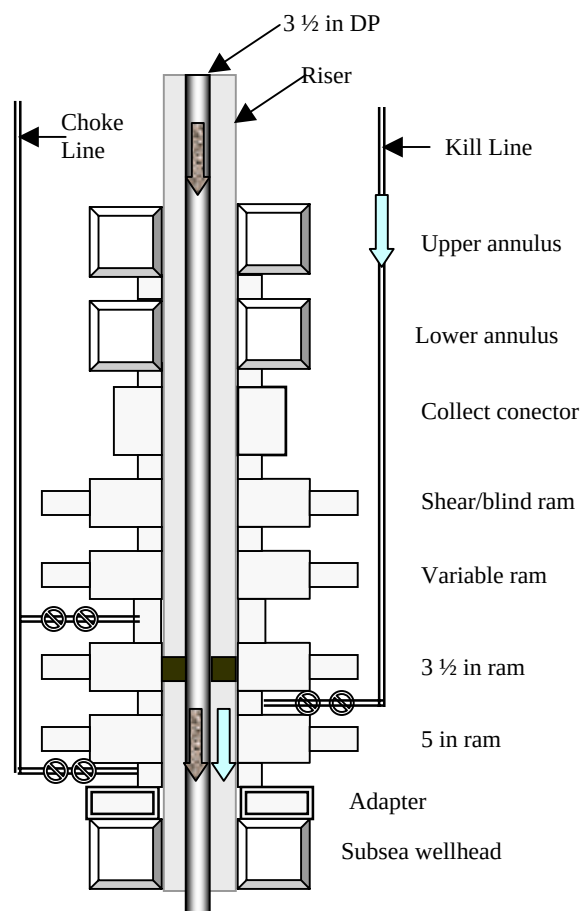


Fig 6. Subsea wellhead and BOP schematics

To keep the hydrajetting tool in position during each fracturing a rubber less packer, working just as a hold down, combined with drill collars and slip joints was used. As the fifth fracture at 3580 m MD was supposed to be a propped one, the rubberless packer was a 9 5/8 in casing one, positioned 1122 m MD above the jetting tool, considering fishing prevention. Cooling shortening was an issue in the long free string below the rubberless packer. This explains the addition of 60/70 mesh sand in the pad and acid stages (Table 2). Experience shows that it is very easy to extend the jetting cuts longitudinally. Another fishing-prevention addition was the placement of 354 m of 3 1/2 in OD flush joint wash pipes immediately after the jetting tool configuration.

Fracturing Design. Designing for hydrajet fracturing follows the same approach as that used for conventional fracturing. Using a commercial simulator and a trial-and-error process, the designers selected the schedule shown in Table 2.

Table 2 – Acidized fracturing schedule

Stage	Clean Vol (gal)	60/70 mesh Sand Conc (lb/gal)	Fluid Type	String Flow Rate (bbl/min)	Annular Flow Rate (bbl/min)
Filling well	20,076	0.00	Linear Gel	8	0
Jetting	5,000	1.00	Gel 40#	17	0 – 4.0
Pad	5,000	0.25	Gel 40#	17	4.5
Acid System	12,000	0.25	Acid	17	4.5
Displacement	4,100	0.00	Linear Gel	17	4.5

The first stage involves filling the well (tubing, annulus and kill/choke lines) with gel. At the end of this stage, the jetting fluid with 1 lb/gal 60/70 mesh reaches the jetting tool. Then the rig closes the choke line valve, and the stimulation vessel initiates the jetting or perforating stage. After 2 min jetting, annular flow starts at a low rate (2 bbl/min); followed by 5000 gal of pad fluid, to open the fracture with simultaneous annulus flow. Next, 12,000 gal of 20% HCl acid (final concentration is about 15% due to annulus fluid flow dilution) is pumped; the acid stage is slightly under flushed into the fracture using 4,100gal of linear gel.

According to the simulation, this schedule should create the fracture shown in Fig 7.

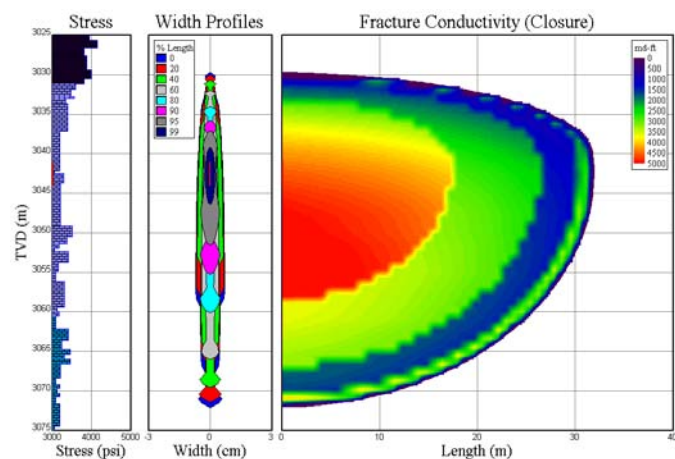


Fig 7. Fracture simulation results

As seen in Fig 7, average fracture conductivity is about 4,000 mD-Ft; with a fracture half-length of 101 ft (31 m) and a fracture height of approximately 101 ft (31 m). This fracture size is acceptable for this job. Note the downward growth tendency of the fracture.

Hydrajet Fracturing Execution. This job, the first stimulation of its kind performed in deep offshore, used real time monitoring through live feed satellite communication. With this technology, experts can see live data worldwide with only a couple seconds time delay and provide on line consultations.

In the following items we present the creation of the five planned fractures at locations 3915 m, 3792 m, 3689 m, 3580 m, and 3435 m MD.

First Fracture (3,915 m). To prepare for the job the rig positioned the jetting tool at 3,915 m MD and rigged up the

fracturing head and treatment lines; the stimulation vessel pressure tested all the system, set relief valves and filled the well with gel, placing the gel with 60/70 mesh sand near to the jetting tool. The rig then closed the choke line at the choke manifold and the stimulation vessel started pumping down the string, through the fracturing head, and down the annulus by the kill line. The job went smoothly (see the job chart in Fig. 8) and according to the plan in Table 2.

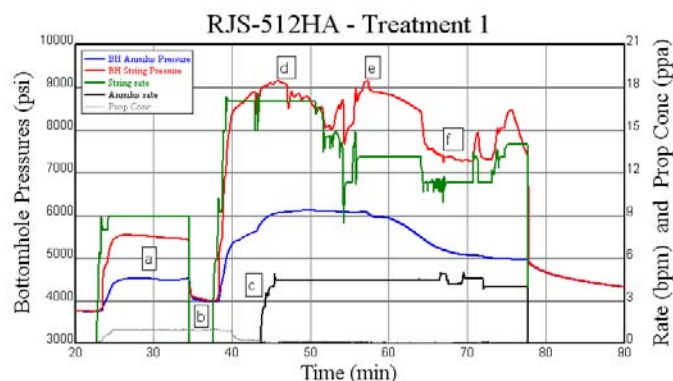


Fig 8. First fracture at 3915 m MD

Keys for the job charts: a) filling the well with gel spotting the gel with 1 ppg 60/70 mesh near the jetting tool; b) jetting start; c) annular flow start; d) pad at formation; e) acid hits the formation; f) acid stage end; displacement start. Note that bottom hole pressures were recorded 1097 m “above” the jetting tool.

Controlling a hydrajet fracturing involves attempting to achieve the planned volumes, rates, pressures and concentrations. The supervisor looks down the tubing for the specified flow rate (17 bbl/min) respecting the maximum pressure allowed. Down the annulus, the pressure must increase until it reaches a stable level, which indicates fracture creation and extension thereafter. However, the annular flow rate is uncertain because it depends on the amount of fluid loss to the areas/fractures previously opened. Fig 8 shows that the annular flow increased until a pressure break occurred (44 min) and stabilized as expected. At about 58 min, a strong drop in annulus pressure was experienced. It was believed that this drop indicated that the fracture had crossed the other lateral, 12 m distant. The treatment followed the acid and flush stages.

Second, Third, and Fourth Fractures (3,792; 3,689; and 3580 m MD). The subsequent three fracture stimulation stages followed the same schedule without major’s problems. Each stage started by moving the tool to the desired location; followed by jet perforation, pad fluid, acid, and flush stages (as previously). Figs 9, 10, and 11 show the job charts.

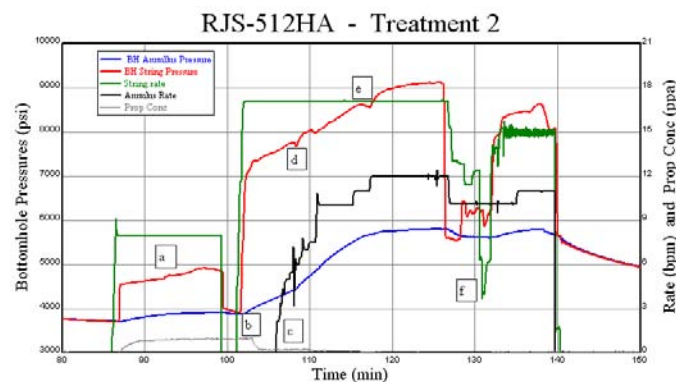


Fig 9. Second fracture at 3,792m MD

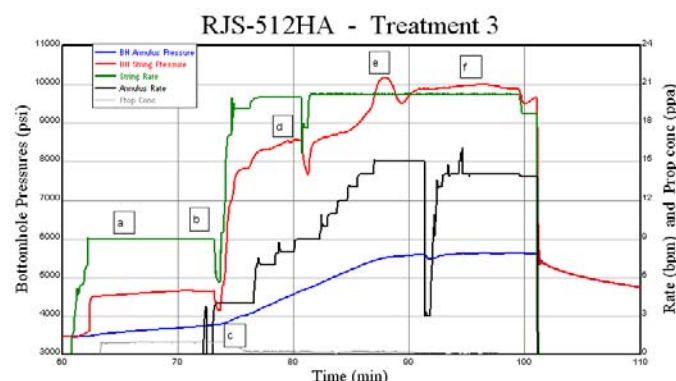


Fig 10. Third fracture at 3,689m MD

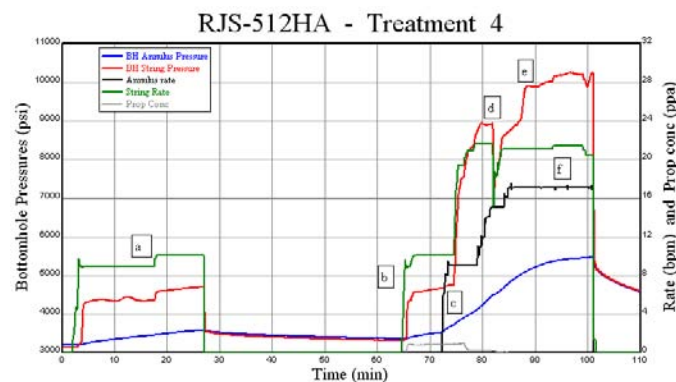


Fig 11. Fourth fracture at 3,580m MD

As the significant variations in the treatments parameters were the annular pressure decrease in the first fracture and the high actual annular rates in the second, third and fourth fractures, Table 3 shows the expected and actual annular rates. It also shows how much it increased between the next fracture and the previous one. Note that the friction loss due to the high annular rate can mask the bottom hole annular pressure behavior, once it was recorded 1097 m “above” the jetting tool.

Table 3 – Annular flow rate (bbbl/min)

Fracture	First	Second	Third	Fourth
Expected annular rate	4,5	6,0	7,5	9,0
Actual annular rate	4,5	12,0	14,5	17,0
Next fracture – previous fracture annular rate	NA	7,5	2,5	2,5

The annular flow rate increased abnormally from the first to the second fracture (from 2.5 bbl/min to 7.5 bbl/min). The fluid loss when filling the well to the second and subsequent fractures was approximately 7 bbl/min. The difference between fractures 2 and 3 and between fracture 3 and 4, is almost the same (2.5 bbl/min), which indicates the contribution of each fracture to the total leakoff. A possible explanation for the excessive leakoff in the second fracture is that the first fracture crossed the 1-RJS-512-HP well, 12 m distant. It is probable that a good acidized fracture connection such as this one allowed a 5 bbl/min fluid leak off into the other lateral; thus exposing the annular fluid to a multilateral system. As shown earlier in Fig. 8, the pressure drop in the annulus during the first fracture confirms that this occurred during that fracture. This phenomenon may have occurred in the second, third and fourth fractures, despite no clear indication in annular pressure behavior.

The high annular rate (above 17 bbl/min) to propagate the fourth fracture prompted the team to replace the planned hydraulic fracturing stage with an acid fracturing stage for the fifth fracture.

Fifth Fracture (3,435 m MD). This last fracture adopted the previous treatment schedule, with tubing rates of 21 bbl/min and annular rate of 18 bbl/min.

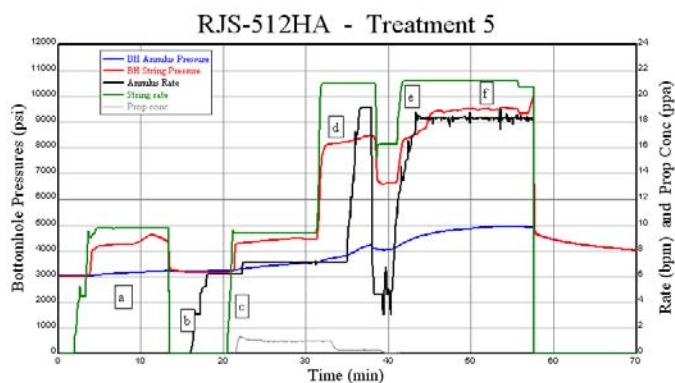


Fig 12. Fifth fracture at 3,435m MD

Results

Focusing on the project's second phase (the subsea well), its operational aspects include the following results.

The hydrajet fracturing parameters followed the simulations, except for the annular rate abnormality explained above. When the first fracture created in the 1-RJS-512HA crossed the other lateral, 12 m distant, it is demonstrated that the fracture direction in Congo field is indeed east-west, like the others carbonates fields nearby.

The hydrajet fracturing string performed well.

Controlling fluid loss when pulling the fracturing string out of the hole and tripping the production tail string in hole by simply feeding fluid at a 7-bbl/hr rate was an easy and safe method.

The recorded temperature variation between the jetting phase and the end of each treatment was approximately 30° F. The respective cooling shortening was about 0.23 m. The addition of 0.5 lb/gal of 60/70 mesh to the pad and acid stages seems to have extended the cuts longitudinally without problems.

The long intervention duration (about 4 months) was due to the associated operations (tubing head recall and exploratory evaluation in a shallower zone) and a long time wait on flowlines connections.

In regard to the oil production after stimulation, the results matched the best expectations. The oil rate sustained at about 305 m3/D for 3 months, (until this written). This rate is three times the rate before stimulation and the predicted decline is a very smooth one, which makes the stimulation a great success (Fig 13). Considering a one-year period from the well shut-in to the intervention in July 2004 (until July 2005), the production balance shows an extra oil production of 224,000 bbl (US\$ 8 million based on \$35-bbl crude). This justifies all the intervention expenditures, because the hydrajet fracturing treatment was a great financial success.

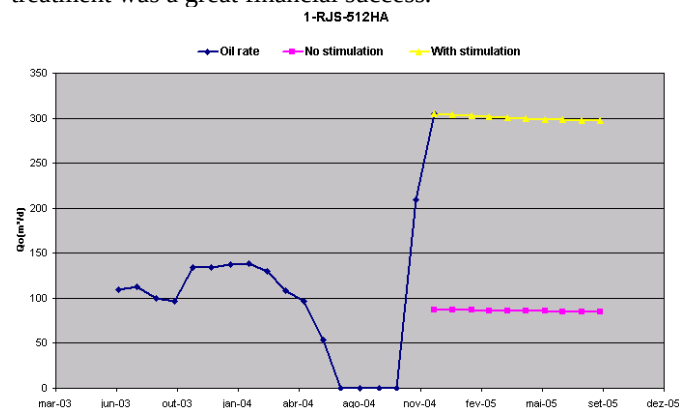


Fig 13. Oil Production history, Well 1-RJS-512HA

Conclusions:

- The good results of the project's first phase (the two wells on fixed platforms), were certainly one catalyst for attempting the hydrajet fracturing process in deep subsea wells. Therefore, attempting new technology implementation onshore or in platform wells proved to be a good approach.
- The project management methods adopted resulted in a very synergistic, pleasant, and productive work environment.
- The operational results of the hydrajet-fracturing treatment in the subsea well, 1-RJS512-HA, made this technology available for all the operator's scenarios—land, platform, and subsea wells.
- Production results obtained from the three wells already hydrajet-fractured indicate that this is a very attractive stimulation technique for horizontal openhole wells.
- The Well 1-RJS512-HA case history proved that hydrajet fracturing is feasible even for multilateral wells after it is possible to deal with the additional fluid loss.
- The hydrajet string used proved to be simple and quite efficient in subsea applications.

Acknowledgements

The authors thank Petróleo Brasileiro S/A, Halliburton Serviços Ltda, and Halliburton for their support and permission to publish this paper. Special thanks go to Maria das Graças F da Costa, João Carlos Neves Calmeto, João

Bosco F Moreira and Márcio Albuquerque Silveira from Petrobras-Campos Basin Unit for their contributions.

Nomenclature

BHA = Bottom hole assembly
 bbl/min = Barrel per minute
 bbl/hr = Barrel per hour
 Cp = Centipoise
 DC = Drill collar
 DP = Drill pipe
 ft = Foot
 ID = Internal diameter
 in = Inch
 lb = Pound
 m³/d = Cubic meters per day
 mD = Mili Darcy
 MD = Measured depth
 kh = Permeability horizontal
 kv = Permeability vertical
 lb/gal = Pounds per gallon
 OD = Outside diameter
 psi = Pounds per square inch
 TVD = True vertical depth

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